

80,000 100 186 2,40,000 17/01/2025 Friday 56 80,000 16
AI is a way of making a computer controlled robot or software or all of them to think it intelligently in the similar way the intelligent human think.

History Perspectives of AI

The idea of AI goes back thousand years. The scientific made the things / objects called as automation that were purely mechanical and without human intelligence.

One of the earliest record of automation comes from 450 BCE which is referred as mechanical pigeon created by a friend of plato.

Many years later one of the famous automation was developed by leonardo da Vince around 1495.

In early 1900's there was a lot of media created which was created around the artificial human.

So, all sort of scientist started to ask "is it possible to create artificial human"?

Some scientist even made some objects products that could be some variant of today's Robot.

The word Robot was first time coined in a chess play 1921 by Karel Capeck Play writer.

1929 Japanese Professor Makoto Nishimura built 1st Japanese robot named as Gakulensaku.

1949 Edmund Callis Barkley, a prob comp. sc. published a book Giant brains or Machine that thinks.

1950 Turing published "computer machinery and Intelligence", who proposed the test of human machine intelligence. that was called limitation game.

1952 Arthen Samuel developed a computer program to play checks the row program that can play game.

1955 John McCarthy held a school / workshop at Dart mouth college in which the term AI is coined.

1958 MacCarthy developed LISP the first programming language for AI

1959 → Arthur Samuel created the word machine learning when giving a speech about teaching machines to play chess.

1960 → 1st industrial robot unimate that started in assembly line at general motors. In New jersey where task was transporting de casting and welded parts of cars which was too dense for human.

1965 → Edward Feigenbaum Joshua Lederberg created the first expert

Search next using AI

20/01/2026 1968 → Alexey Ivakhnuk USSR Mathematician published a paper 'Group methods of data handling which proposed a new approach for AI.

no of methods < no of variables

→ Linear Regression -

$$E = \sum (T_i - T_o)^2$$

$$\frac{\partial E}{\partial T_o} = -2 \sum (T_i - T_o) = 0$$

$$\sum_{i=1}^n T_i = \sum_{i=1}^n T_o = 0$$

1200 BCE

$$F_{n+2} = F_{n+1} + F_n$$

Pingala

Polyanshi

$$h^2 = p^2 + b^2$$

Badhyani - Pythagoras theorem

PAP theorem

$$\sum_{i=1}^n T_i = T_o \sum_{i=1}^n 1$$

100 Cr = 1 Billion

10 Lac = 1 Million

$$T_o = \frac{\sum_{i=1}^n T_i}{n}$$

→ Algol → 60

→ 1973 → James Lighthill made a report to British Science Council underlying that stats were not impressed as promised by scientists.

→ Scientists which led to cut in funding.

→ 1979 → James Adam developed a stanford cart which was the first autonomous vehicle. It could navigate a room full of chairs without intervention of human.

In some year Association of American Artificial Intelligence (AAAI) was established.

→ 1980 - The 1st conference of AAAI was held at stanford at 1980. In this conference 1st expert system.

come into market. XCON. (Expert confuser)
It was designed to assist in ordering to the computer systems by automatically picking the components based on the customer's needed.

→ 1981 → Japanese govt allocated \$800 for developing 5th generation computers with aim to create a computer that can translate converse in human language and explain reasoning at human level.

→ 1986 → An autonomous drawing program AARON was demonstrated at AAAI Conf.

Ernst Dickmann and his team created and demonstrated the first driverless car (Robot car) that could drive upto 55 mph on roads.

→ 1987-93 → AI winter.

1987 → the market of specialized LISP and hardware collapsed due to cheaper and more common competitors that could run LISP like IBM, APPLE

1997 → Deep Blue (IBM) beat the world chess champion. Gary Kasparov in a highly publicized match. Becoming the first program to beat human in chess.

21/01/2026

Taylor Series → $f(x)$ can be expanded

$$x_{n+1} = x_n + \theta (x_n - x_{n+1})$$

$$x_{n+1} = x_n + \theta \Delta x_n$$

input x_n output y_n
 $x_1 \quad x_2 \quad x_3$

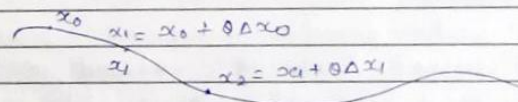
$$w_{i \text{ new}} = w_{i \text{ old}} + \theta \Delta w_i$$

$$y = 1 \quad \text{if } \sum w_i x_i > T$$

$$y = 0 \quad \text{if } \sum w_i x_i \leq T$$

Activation Function

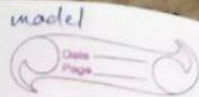
Error = $|Y_{\text{actual}} - Y_{\text{computed}}|$



→ always take $\theta < 1$ because if we take θ having greater value than the value along the curve moves very fast than some conditions on the curve may be skipped.

→ Neural Networks and Deep learning

→ GPU / Machine Learning



- 2000 → Prof Cynthia Breazeal developed the first robot that can simulate human emotions with its face which includes eye, eyebrow, ears, mouth, kismet.
- 2002 → The first roomba it can clean any type of floor
- 2010 → Microsoft launches XBox, the first gaming hardware designed to track body movement and translate it into gaming direction.
- 2011 → An NLP computer programmed to answer questions, Watson by IBM, it was jeopardy quiz by defeated two champions Brad Rutter and Ken Jennings.
- 2012 → Two researchers from Google Jeff Dean and Andrew Ng, trained a neural network to recognize cats without giving labelled data.
- 2016 → Hanson robotics created a humanoid robot named Sophia, who became the first robot citizen. which was created with a realistic human appearance. and the ability to see and replicate emotion and communication.
- 2017 → Facebook
- 2018 →
- 2019 → Google Alphabets.
- 2020 →
- 2021 → Open AI.

AI → Artificial general intelligence

23/01/2025

AI [Artificial Intelligence]

Agents:- An agent is anything that can be viewed as perceiving its environment through sensors and eyes ears acting upon the environment through actuators hands and legs.

We need agent to carry out a task.

→ The agent function: f set of percepts → set of action
 $f: P \rightarrow A$

→ The agent program runs on the physical actuators to produce b.

$$\text{agent} = \text{ac} + \text{program}$$

→ Initial state $[P_1]$
 (Some problem is given)

performing some operation

Result

$[P_2]$ Initial state

take random inputs

$$A = \{2x \mid x \in I\} \quad B = \{x \mid x \in I \text{ and } x \text{ is divisible by } 2\}$$

$$A = P_1 \quad B = P_2$$

* Vacuum Cleaner :-

Percept sequence	Action
[A. clean]	Right
[A. Dirt]	Suck up
[B. Dirt]	Suck up
[B. clean]	Left.

percepts location, contents [A. dirt]
 Action left, right, suckup, Noop

* Rationality :- It refers to ability of an AI agent to take decisions that maximize its performance based on the information it has and the goal it seeks to achieve.

→ A rational AI system aims to choose the best possible actions from a set of alternatives to achieve a specific objective. This requires logical reasoning, learning from experience and adapting to new situation.

27/10/2025 Monday

Environment → (i) Deterministic
 (World) → (ii)
 the environment can be of various types.

- (i) Fully observable vs partially observable
- (ii) Deterministic vs Stochastic
- (iii) Episodic vs Sequential
- (iv) Static vs Dynamic
- (v) Discrete vs Continuous
- (vi) Single agent vs Multi agent.

Q. How to measure the information?
Q. measure of uncertainty?

No. of possible moves in a chess. = 10^{120}
 In Go no. of possible moves = 10^{1720}

→ Stochastic → probabilistic [In environment at each step there are multiple moves.]

(i) * Fully observable vs partially observable
 If everything an agent requires to choose its action is available to it via resources, it is called fully observable.

If parts of the environment is not available then it is called partially observable.

* Taxi driving → partial
 * part picking → fully
 * Image analysis →

Color image - It contains three matrix with colors.
(RGB) $\rightarrow$ Red, green, Blue.

Video $\rightarrow$ Collection of frames.
16 frames/sec
25 frames/sec
30 frames/sec.

* Frequency of landline 30Hz to 3400Hz

$\rightarrow$ Deterministic v/s Stochastic

If the change of states depends on the current state and agent action, then it is called deterministic environment.

ex:- Image analysis

If some of state are beyond the control of the agent, then it is called stochastic, non-deterministic, probabilistic.

ex:- taxi driving.

28/01/2026

$\rightarrow x(t_1, x_1) - x(t_2, x_2) = 0$

iff $x_1 = x_2$ and $t_1 = t_2$

$\rightarrow$ Episodic and Sequential

$\rightarrow$ If choice of current action is depend on previous action then the environment is said to be episodic

does not

If it depends then environment is called sequential environment.

in this case the present action decides the future action.

taxi driver $\rightarrow$ Sequential

Part picking
Image analysis } Episodic

$\rightarrow$ Static and dynamic

$\rightarrow$ The environment does not change with the passage of time.

$\rightarrow$ In dynamic there is a change with time.

$\rightarrow$ Agent can plan before hand what to do.

$\rightarrow$ while making plan / taking decision we need to take into consideration that change.

Semidynamic $\rightarrow$ in this environment does not change with time but the performance scene of the agent is change.

Taxi $\rightarrow$ dynamic
Poker $\rightarrow$ Static

Fourier transformation $\rightarrow$ any function can be represented in the form of $\sin$ and $\cosine$.

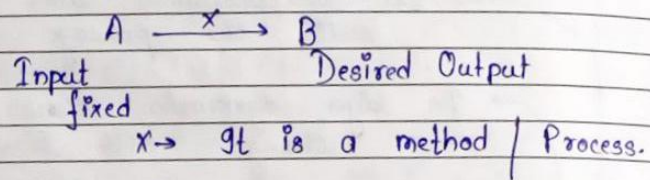
$\rightarrow$ Discrete and Continuous $\rightarrow$

If there are a limited number of distinct clearly defined percepts and actions then it is discrete.

If there is a range of value then it is continuous.

$$f \in [a, b]$$

$\rightarrow$ Single Agent and Multiagent $\rightarrow$



$\rightarrow$ Environment Types :-

$\rightarrow$ Agent Types $\rightarrow$

There are four basic types in order of increasing generality.

- ① Simple Reflex agent
- ② Reflex agent with state
- ③ Goal based agent
- ④ Utility based agent.

06/02/2025

$\rightarrow$ Utility based agent $\rightarrow$ the goals are not always enough for example many action sequence get the task to destination.

Things like how fast, how cheap, how safe.

$\rightarrow$ An utility function maps a state onto a real number that describes the associated degree of success / goodness.

$\rightarrow$ A complete specification of utility function allows a rational decision.

$\rightarrow$ In two kind of situation where goals are inadequate

$\rightarrow$ first when there are conflicting goals. only some of them can be achieved. the utility function specifies the appropriate trade off

safety speed
$$f(s, s) = \alpha \text{ safety} + \beta \text{ speed}$$

$\alpha + \beta = 1$

Q. Why we need normalization?

$\rightarrow$ Second when there are several goals that the agents aim for none of them can be fulfilled with certainty.

→ the utility function provides away in which the likelihood of success can be weighted up against the importance of goal.

→ An agent can be turned to a learning agent.
→ An agent program can select action using different methods

→ The point is how the agent program come into being

Turing in 1950 considers the idea of actually programming his intelligent machine by hand. He estimate how much work it might take.

The method he proposed is to build learning machine and then teach them.

→ Learning allows an agent to operate in initially unknown environment and to become more competent than it was initially.

→ A learning agent can be divided into four conceptual component

- ① learning element
- ② performance element
- ③ critic
- ④ problem generator

① the learning element is responsible for making improvement.

② The performance element is responsible for selecting the external actions. It takes in percepts and decides on action.

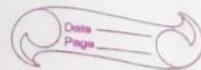
the learning element uses feedback from critic and how the agent doing determines how the performance element should modified to do in future.

→ The design of learning element depend upon the design of performance element.

when trying to design an element the critic tells the learning element how well the agent doing with respect to a fixed performance standard conceptually the performance standard

10/02/2024

Monday



→ Problem Solving :-

① goal formulation - based on current situation and agent prob performance.

② Problem formulation → It is a process of deciding what action and state to be considered for the given goal. immediate

→ An agent with several options of unknown values can decide what to do first by examining different possible sequence of actions that leads to state of known value and then choosing the best sequence.

→ formally a problem may be defined

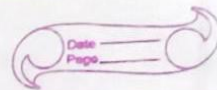
① Initial state → It is the state when agent is in.

② Set of possible actions it is done by successor function.

③ Goal state

④ Evaluate / cost of path. (Solution)
Utility function

→ The most common formulation uses the successor function. Given a particular state x , the successor function takes x as input and returns a set of (action, successor) which is an ordered pair whose each action is one of the legal actions in the state, and each successor is a state that can be searched from x by applying actions.



→ the initial state and successor function implicitly define the state space of the problem, which consists of all states which are reachable from initial state.

→ the state space can be pictured using graph in which nodes represent the state and the arch between the nodes represent the nodes.

→ A (path) in the state space is always a sequence of states connected by a sequence of actions.

→ Goal test It determines whether a given state is goal state. Sometimes there is an explicit set of possible goal states and the test simply checks if the given state is one of them.

→ Path cost → It is function that assigns a numeric value to each path. the problem solving agent chooses a cost function that reflects it's own performance measure. the cost of path can be described as the sum of cost of individual actions. along the path.

→ A solution to a problem a path with minimum cost.

→ Problem solving in AI → It is the core concept of AI that mimics human cognitive. It involves identifying the challenges, analyzing situation and applying state sequence to find effective solutions.

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The problem solving agents analyse the situation and employ the various techniques to achieve the desired outcomes.

11/02/2025 Steps of problem solving-

- ① state $\rightarrow$ (state space) $\rightarrow$ It consists of all possible states.
- ② Initial state $\rightarrow$ one of the states is considered as the initial state depending on the problem.
- ③ Successor function $\rightarrow$ It involves the functions associated with the selecting state and identifying all possible actions. Each action generates a new state.
- ④ Goal test $\rightarrow$ This checks if the new state generated is a goal state.
- ⑤ Path cost $\rightarrow$

$\rightarrow$ the best solution/path is the path having optimal cost.

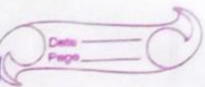
8 Puzzle Game $\rightarrow$ a square having 9 blocks with 8 disks written 1 to 8 on it and have to be arranged in sorted manner.

$\rightarrow$ 8 Queen Problem $\rightarrow$

$\rightarrow$ Vacuum cleaner $\rightarrow$ possible states

17/02/2025

Monday



8-queen problem / puzzle problem / Tower of hanoi

States $\rightarrow$ Combination of poles and disks.

Initial state $\rightarrow$ Pole A has all the disks pole B and C are empty.

Successor function $\rightarrow$ Move disk one pole to another

Subject to $\rightarrow$ ① higher disk can not be put on smaller disk.

② In one move only one disk can move.

③ If there is a disk under a disk or any number of disk then that disk can not be moved.

Rubik cube $\rightarrow$

States $\rightarrow$ All possible configuration with different color on a single face.

Initial state $\rightarrow$ one of the states

Successor function $\rightarrow$ Rotate left right up-down front-back.

Goal test $\rightarrow$ Each face should have a single colour.

Path cost $\rightarrow$ 1 per rotation.

To solve $\rightarrow$ AI problem search is main component.

★# There are two types of Search

- ① Informed Search
- ② Uninformed Search.

→ An uninformed Search is also called blind method. don't have any additional information other than the problem definition about goal / path to reach the goal.

They explore to reach the space without any guidance relying purely on the structure of the problem.

Methods → Breadth first ~~search~~ Search
depth first ~~reach~~ Search
uniform / simple cost / ~~reach~~ Search

These methods are simple to implement and guidance to find the goal (if it is there) they can be inefficient in terms of time and space.

→ Informed Search → They are also known as heuristic search algorithm. They have additional information to guide the search process.

this additional info helps to estimate the cost / distance to the goal. Allowing the algorithm to choose certain path over others.
Common Methods A*, greedy, best first search.

They are more efficient as they significantly reduce the no of nodes explored. They may not always guarantee the optimal solution.

- Complexity = $O(b^{d+1})$ (BFS)
- Fringe = all the visited nodes other than leaf nodes.

Function BFS (problem fringe)
fringe insearch (maknode [initial state [Problem] fringe])
1st loop → if fringe is empty return failing
node ← Remaining node → First (fringe)
if goal test (problem) applied to the state node successor then return solve state nodes.
fringe ← insert All (expand (node problem) fringe);

18 Feb 2025

DFS (depth first search) → Stack (LIFO)
Depth limit Search

Completeness → if there is a solution / goal node it also should find it.

Optimality → if it finds the solution it should give best path.

Incremental Deepening Depth Search

Uniform Cost Search →

If the cost of each step is different and all are positive.

We need to choose the node for expansion that has the minimum cost.

Informed Search techniques → These methods require some information other than the problem definition. This information helps to estimate the path with minimum cost without

exhaustively finding the cost. This is generally given in terms of heuristic function.

A heuristic function is used to estimate the cost of reaching a goal from initial state. It guides the search for a solution and provides a way to access the desirability of different options without exhaustively exploring every possibility. It helps to prioritize the exploration of more promising path.

Greedy Best First Search →

Bidirectional Criteria	BFS	Uniform	DFS	Depth	Incremental
$y_{a,d}$ Completeness	y_a	$y_{a,b}$	N	Limit N	y_a
$O(b^{d/2})$ Optimality Time	$O(b^{d+1})$	$O(b^{c \cdot l})$	$O(b^m)$	$O(b^l)$	$O(b^d)$
" Space	$O(b^{d+1})$	"	$O(b^m)$	"	$O(b^d)$
$y_{c,d}$ optimality	Y	Y	N	N	Y ^c

B- branching factor
m- depth of tree
l- limit of depth

y_a = complete if b is finite.

y_b = complete if step size $\epsilon > 0$

c = optimal if step cost is identical

d = if best decision are BFS

c^* is cost of optimal path.

20/02/2025

Informed Search → In this we have additional information other than the problem function

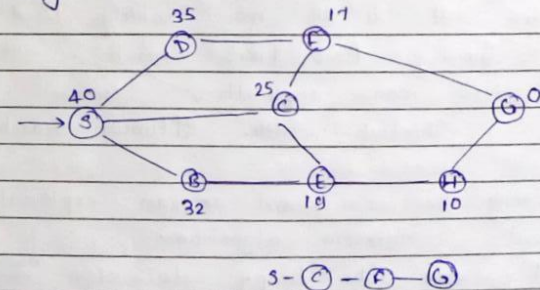
Heuristic

Estimate of distance from a node to goal.

Greedy Best first Search → $f(n) = g(n) + h(n)$ s → current node distance from current state to goal state (Exact).

Dijkstra Algorithm →

Greedy Best first Search →



greedy Best first Search.

Greedy Best First Search

- It is easy to implement, fast, requires low memory.
 - It is not always guaranteed to optimal path.
 - It may get stuck in local minima.
 - It lacks completeness.
 - It can be used in Games, NLP, ML.
 - It resembles the best first search. In the way it refers to follow a single path. all the way to goal, but will back up when it hits a dead end.
 - It suffers from the same problem as the depth first search.
- Worst time complexity $O(b^m)$.

Q. Is binary tree a tree?

Ans. Need not be.

- ① Empty set can be a binary tree.
- ② A node can have at most 2 children.

Q. A* Search → It is the most widely used form of Greedy Best First Search. It is used in many application - Games Finding, Shortest Path, Optimum Search, Robots.

- It may be considered as an explanation of dijkstra algorithm.
- It uses the heap to store ^{nodes} search.
- It combines the benefits of dijkstra and greedy best first search.

→ It uses the heuristic function to evaluate the path by ^{balancing} the cost to search reach the current node. and the cost from current node to goal node.

→ It ensures fewer paths than Dijkstra algo while generating the shortest path. provided the heuristic function is consistent

never over estimates the cost along the

the consistent heuristic for every n and successor m $h(n) \leq \text{cost of } n \text{ to } m \text{ through } \text{successor } m + h(m)$

24/02/2025

A* algorithm →

how to calculate heuristic function value.

1. Deterministic → we need to have to cost between each pair of nodes which is quite costly in terms of time as well as money.

2. Estimation → there are three ways

- ① Manhattan Distance
- ② Diagonal Distance
- ③ Euclidean Distance

→ Manhattan Distance.

→ Diagonal Distance →

$$dx = \text{abs}(\text{current } x - \text{goal } x)$$

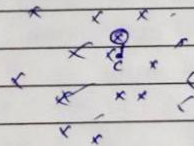
$$dy = \text{abs}(\text{current } y - \text{goal } y)$$

$$h = D \times (dx + dy) + (D_2 - 2D) \times \min(dx, dy)$$

where D = length of every node and it's default value.

D_2 = length of diagonal of every node and it's default value $\sqrt{2}$.

It should be used when there are eight decisions of movement.



$$d((1, x)) < D$$

$$\bar{x} = (x_1, y_1) \quad \bar{y} = (x_2, y_2)$$

$$(x_2 - x_1)^2 + (y_2 - y_1)^2$$

$$d(\bar{x}, \bar{y}) = (\bar{x} - \bar{y})(\bar{x} - \bar{y})^T = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

$$d_2(\bar{x}, \bar{y}) = (\bar{x} - \bar{y}) \& (\bar{x} - \bar{y})^T \neq (x_2 - x_1)^2 + (y_2 - y_1)^2$$

↳ cor-relational matrix

1940

71 A^* → It maintains two lists OPEN and closed.

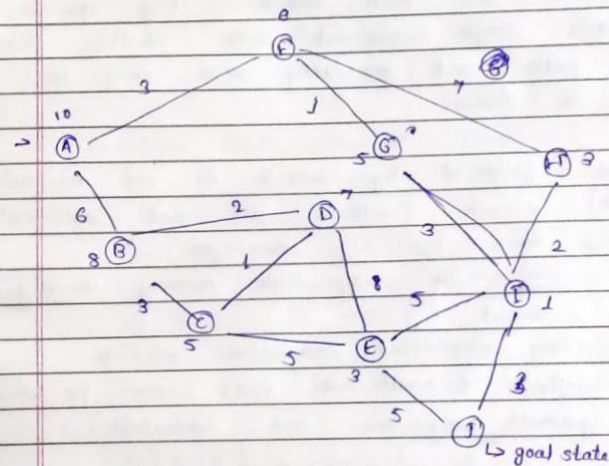
s in open = $\{B\}$

s in close = $\{S\}$

put all successor of s in to open.

$$f(n) = g(n) + h(n)$$

- ① Initially open contains the initial states S .
- ② If open is empty return failure.
- ③ Remove a node with minimum cost from open and move it to closed.
- ④ if n is a goal state return success.
else expand n (find successor of n).
put all successor of n into OPEN.



25/02/2025

the best first search and A^* algorithm are designed to explore the search space systematically.

The systematically is achieved by keeping one or more paths in memory and by recording which alternatives

have been explored at each point along the path and which have not been. In many application the path to goal is irrelevant eg IC design.

Job shop scheduling, vehicular Routing if the path is irrelevant, we may have a different class of algorithm which are generally called as local search. They operate using a single current state rather than multiple path and generally move only to neighbour to states.

The path followed by search is not maintained. the local search methods are not systematic but they have two key advantages.

① they use very little memory usually a constant amount

② They often find reasonable solution in large infinite (continuous) state space for which the systematic algorithm are unsuitable.

In addition to find the goal local search algorithm are useful for solving optimization problems.

To understand the local search we find it very useful consider the state space landscape which has both location (defined as state) and the elevation (defined by the value of heuristic function or objective function),

the local search algorithm explore landscape.

1) Hill climbing algorithm :- It continuously moves in the direction of increasing value and terminates when it reaches the peak (no higher values)

→ It maintains a search tree.

→ It only maintains state and objective function value.

→ It does not look ahead beyond the immediate algorithm of current state.

function hill climbing (problem) {

current = ~~metanode~~ metanode (initial state [problem])

loop

neighbour → highest value successor of current

if value [neighbour] < value [current]

current → neighbour.

IS

3	1	4
5	2	7
6	8	

GS

1	2	3
4	5	6
7	8	

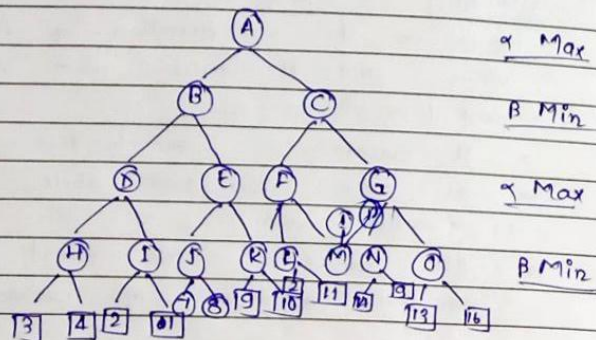
$$\text{cost} = |2-1| + |5-2| + |3-1| + |4-3| + |5-4| + |7-6| + |8-7|$$

$$= 1+3+2+1+1+1+1=9$$

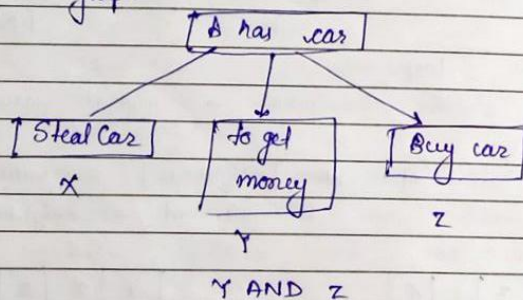
03 - March - 2025



Min-Max [α-B Pruning algorithm] [Pruning Algorithm]



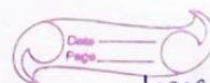
AND - OR graph



AO* search → It does not explore all nodes.

- we initialize the graph to start node.
- Traverse the graph following the current path
- accumulating nodes that have not yet been expanded.
- Pick any of those node and expand it and if it has no successor then $f(n) = \text{utility}$.

04 - March 2025



Logic → It is a greek word which has different meanings, sentences, reason discourse, rule and ratio

It is structure of current reasoning.

the simplest logic is propositioned logic. which consist of sentences

A sentence in logic has two values only true and false.

→ Well formed formula

1. Constants are wff.
2. P is a wff
3. if P and Q wff then $P \vee Q$ are wff and $P \wedge Q$ are wff $\neg P$ is wff $P \Rightarrow Q$ is a wff (P) is a wff

< formula > Atomic proposition

→ It is a proposition whose truthness is not depend upon any other proposition

< formula > = atomic propositions | - < formula > | < formula > $\vee$ < formula > | < formula > $\wedge$ < formula > | < formula >

! < formula > $\Rightarrow$ < formula > | < formula > $\Leftrightarrow$ < formula > | < formula > | T/F |

L7

06-03-2025

Logic

Proposition logic
Predicate logic

basic component

Constant - true or false

Variables -

AS is taught in MCAS year
 $F(x_1, x_2) = x_1 \wedge x_2$ T T $\in M$ T F $\in M$ F T $\in M$

F F

Predicate logic $\rightarrow$

AS is taught in MCAS year.

x is taught in y.

x and y are variables.

 $\exists, \forall$ operator in addition to what we have in propositional logic. $\forall \wedge \neg \Rightarrow \Leftarrow$ $\forall y \forall x (\text{is taught}(y, x))$ $\downarrow$ Some list of program.

Some list of subjects

Skolemization those constants are called skolem constants

for every false $\hookrightarrow$
x it is true. $\forall x P(x) \rightarrow$ is true in a model. $\exists x P$ is true in a model m iff P is true with x being some possible object is m. $\forall x P(x) \Leftrightarrow \neg \exists x \neg P(x)$ $\hookrightarrow$ P is always false.~~OR has less complexity than AND operator.~~ $\exists x P \Leftrightarrow \neg \forall x \neg P(x)$

01-March-2025 [Friday]

 $\rightarrow$ Any formal system can be considered a logic system if it has

- $\rightarrow$ A well defined syntax - rules
- a well defined semantics $\rightarrow$ related meaning
- a well defined proof theory $\rightarrow$ derive some ~~or~~ new thing from existing

Everyone likes icecream

 $\forall x \text{ likes}(x, \text{icecream})$ $\exists x \text{ likes}(x, \text{icecream})$

Someone likes icecream

Nobody dislikes icecream

 $\neg \exists x \neg \text{likes}(x, \text{icecream})$ $\& \neg \forall x \neg \text{likes}(x, \text{icecream})$

$$A \Leftrightarrow B \equiv (A \Rightarrow B) \text{ AND } (B \Rightarrow A)$$

$X = (\neg A \vee B) \wedge (B \vee A) \Rightarrow T$ if both are either true or false.

XNOR $\rightarrow$

A	$\neg A$	B	$\neg B$	X
T	F	T	F	$\rightarrow T$
T	F	F	T	$\rightarrow F$
F	T	T	F	$\rightarrow F$
F	T	F	T	$\rightarrow T$

$A \models B$

A entails B.

B is entailed by A.

true value of B is subset of true value of A.

if A is true then B will be definitely true.
if B is true then A may or may not be true.

Modus Ponens $\rightarrow$

$$\begin{array}{c} A \rightarrow B \\ A \\ \hline B \end{array}$$

Modus tollens

$$\begin{array}{c} \neg A \rightarrow \neg B \\ \neg B \\ \hline \neg A \end{array}$$

Programming language with logic.
Prolog, Pylog

$\rightarrow$ All students of a philosopher read one of their teachers book.

$\forall x \forall y [\text{philosopher}(x) \wedge \text{students of}(y, x) \rightarrow \exists z [\text{Book}(z) \wedge \text{write}(x, z) \wedge \text{Read}(y, z)]]$

$\rightarrow$ Skolemization $\rightarrow$

$\forall x \forall y [\text{philosopher}(\text{Hazi}) \wedge \text{student}(\text{Rom}, \text{Hazi}) \rightarrow \exists z [\text{Book}(AI) \wedge \text{write}(\text{Hazi}, AI) \wedge \text{Read}(\text{Rom}, AI)]]$

Change any variable with constant or all variable with constant is called skolemization.

$\rightarrow K B$

$$\begin{array}{l|l} & (\neg D \vee A) \vee D = A \\ R_1 : (A \vee \neg C) \Rightarrow B & \neg(A \vee \neg C) \vee B \\ R_2 : (\neg C \vee D) & (\neg A \wedge C) \vee B \\ R_3 : D \Rightarrow A & \text{false} \\ R_4 : C & \\ R_5 : D & \\ R_6 : A & \\ R_7 : B & \end{array}$$

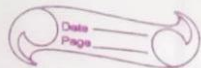
$$\begin{array}{l} R_1 : (A \vee B) \wedge C \Rightarrow E \\ R_2 : D \wedge F \\ R_3 : \neg E \vee D \\ R_4 : B \wedge C \rightarrow \text{And both are true} \\ R_5 : F \\ R_6 : B \\ R_7 : D \\ R_8 : E \\ R_9 : D \\ R_{10} : A \end{array}$$

$$\neg((A \vee B) \wedge C) \vee E = ((\neg A \wedge \neg B) \vee \neg C) \vee E$$

$$\neg(D \wedge F) \vee A \vee E$$

$$((\neg A \vee \neg C) \wedge (\neg B \vee \neg C)) \vee E$$

11-March-2025



$F = \{A, A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow E, E \rightarrow F, F \rightarrow G\}$
 $\rightarrow$ Does F entail $\{F \vee G\} \rightarrow F \vee G$

$A \models B$

A entails B if whenever A is true
 then B must be true.
 B is entailed by A .

$A \rightarrow B$	$\equiv$	$\neg A \vee B$
① A is true		
② B is true		

According to this all literals
 of F are true.
 so $F \vee G$ is true

- ① Conjunctive normal form
- ② Disjunctive normal form

$$\bigvee_{i=1}^n \left(\bigwedge_{j=1}^m L_{ij} \right)$$

$$\bigwedge_{i=1}^n \left(\bigvee_{j=1}^m L_{ij} \right)$$

$\rightarrow$ Any logical formula can be converted into form
 of CNF and DNF.

$\rightarrow$ Principal CNF

Min term

Max term

$$P \wedge Q, \neg P \wedge Q, P \wedge \neg Q, \neg P \wedge \neg Q$$

$$P \vee Q, \neg P \vee Q, P \vee \neg Q, \neg P \vee \neg Q$$

Principal DNF

11 day March April May

$$(A \vee B) \wedge (\neg C \vee D)$$

$$(A \vee B) \wedge (C \wedge D) \wedge (\neg A \vee \neg B \vee \neg D)$$

$$\text{CNF} = \wedge(VL)$$

$$\text{DNF} = \vee(LN)$$

$$P \wedge (P \rightarrow Q)$$

$$P \wedge (\neg P \vee Q)$$

$$(P \wedge \neg P) \vee (P \wedge Q)$$

$$P \wedge Q$$

Horn formula $\rightarrow$ A CNF is called Horn formula
 if each conjunction contains atleast
 one positive literal.

$$F = \bigwedge_{i=1}^m \left(\bigvee_{j=1}^n L_{ij} \right) \text{ where } L_{ij} \text{ are literals}$$

and for each i there
 is not j with L_{ij} positive.

Thursday

20-March-2025

$\rightarrow$ In any formal system,

we have a system of axioms which
 are sufficient enough to expand basic
 concepts required for the system.

and set of function predicates
 that use the axioms and generate new facts/
 objects.

Algol $\rightarrow$ Algorithm language.

Syntactic $\rightarrow$ [Syntax]

Sugar

Syntactic Salt $\rightarrow$

Knowledge Base

- $\rightarrow$ It is a set of sentences.
- $\rightarrow$ It can be extended and queried by two standards.
Name $\rightarrow$ Tell and Ask.
- $\rightarrow$ The knowledge base agent program maintains KB, which may initially contains some basic background facts. Each time the agent program is called it does two things. -

Function KB agent program [percept]

```
Static : KB
t : time initialized to 0.
Tell (KB, Make percept Sentence (percept, t))
Action  $\leftarrow$  Ask (KB, Make - action - query (t))
Tell (KB, Make - action - Sentence (Action, t))
t  $\leftarrow$  t + 1
return action.
```

- $\rightarrow$ first, it tell KB what it perceives
- Second, it asks KB what action it should perform. Once the action is chosen, it records the choice and execute the action.
- $\rightarrow$ The second tell is needed to let the KB know that the action has been actually performed.

Ex $\rightarrow$ Rohit has exactly two brothers.
 $\exists x, y$ brother (x, Rohit) brother (y, Rohit)
 $\wedge (x \neq y)$
if $x = y = \text{Mohan}$ Brother (Rohit) = Mohan

- $\rightarrow$ we can add statement / assertion in KB.
eg $\rightarrow$ Rohit is a king and kings are person
Tell (KB, King (Rohit))
Tell (KB, $\forall x$ King(x) $\Rightarrow$ Person(x))

24/03/2025

Formal Systems

[Oxioms which are facts assumed to be true]

Oxioms

$$P(\emptyset) = 0 \quad P(S) = 1$$
$$P(A) \leq P(B) \quad \text{if } A \leq B$$

Sets $\rightarrow$ [Oxioms]

- $\rightarrow$ 1. the only sets are empty set and those made by adjoining something to a set.

$$\forall s \text{ Set}(s) \Leftrightarrow (s = \{\}) \vee (\exists x, s_2 \text{ Set}(s_2) \wedge s = \{x\} \cup s_2)$$

- 2. the empty set that has no element that were adjoined into it that is there is no way to decompose empty set into smaller set or in an element.

$$\neg \exists x \ S \ \{ \text{add}(x, S) \} = \{ \}$$

3. Adjoining an element already in the set has no effect.

$$\forall x \ S \ x \in S \Leftrightarrow S = \{ \text{add}(x, S) \}$$

4. the only members of a set are the elements that were adjoined into it. it can be ~~expressed~~ ^{expressed} recursively by saying that x is a member of S iff S is equal to some set S_2 adjoined with some element y where y is the same as x or x is a member of S_2 .

$$\forall x, S \ x \in S \Leftrightarrow [\exists y, S_2 (S = \{ \text{add}(y, S_2) \} \wedge (x = y \vee x \in S_2))]$$

5. A set is a subset of another set iff all the elements of the first set are the members of second set.

$$\forall S_1, S_2 \ S_1 \subseteq S_2 \Leftrightarrow (\forall x \ x \in S_1 \Rightarrow x \in S_2)$$

6. Two sets are equal iff they are subset of each other

$$\forall S_1, S_2 \ S_1 = S_2 \Leftrightarrow (S_1 \subseteq S_2 \wedge S_2 \subseteq S_1)$$

7. An object is in intersection of two sets iff it is a member of both sets.

$$\forall x \ S_1, S_2 \ x \in S_1 \cap S_2 \Leftrightarrow (x \in S_1 \wedge x \in S_2)$$

8. An object is in the union of two sets iff it is member of any of the two sets.

$$\forall x \ S_1, S_2 \ x \in (S_1 \cup S_2) \Leftrightarrow (x \in S_1 \vee x \in S_2)$$

Knowledge Representation

When we are the search to solve a problem, we must capture the knowledge needed to formulate the problem.

Apply a search technique to solve the problem execute the problem solutions.

"Can computer solve a problem"?
 → which is solvable.

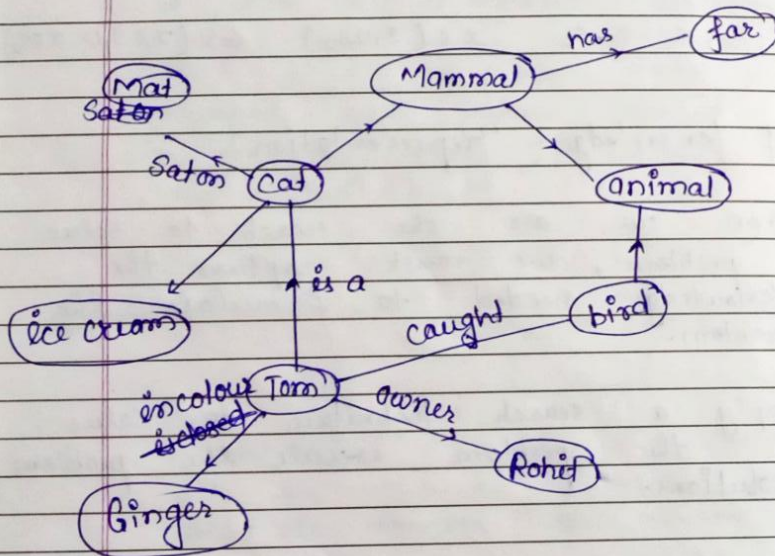
03/04/2025

★ ★ Knowledge Representation →

The semantic nets solve the knowledge in the form of graph

Logical Representation
Semantic Nets
frames

the nodes in the graph represent objects entities and links represent the relationship between the corresponding nodes



Tom is a cat.
Tom is owned by Rohit
Tom is ginger in color
Cats like ice cream
Cat sat on mat
A cat is a mammal

Relation → A subset of cartesian product.

A bird is an animal
All mammals are animals.
Mammals have fur
Tom captures bird.

the form of representation is closer to human structure knowledge by building mental links between objects than the logic. All information about an object is concentrated on the node representing that node.

there are some confusion that stems from the imprecise nature of the semantic nets.

we can not differentiate between an individual and a class.

One more limitation is that the link between the objects represents only binary relations.

eg:- Run (Vande Bharat, Delhi, Varanasi)

frames → It provides a more representation of knowledge. as compared to semantic nets.

In place of an arbitrary no of arcs. loading from a node.

there is a fixed number of slots representing the attributes of an object.

Every object is a member or instance of a class.

15/04/2025

Conceptual dependency :- It is the theory that is used to represent the knowledge / sentence in a neutral language independent of the language.

Independent of the structure

EX → Hari gave a book Ram.

Ram was given a book by Hari.

They are different structurally but same conceptually.

Conceptualization represents a specific event or action composed of an actor or any cases depending upon that action.

→ It focuses on understanding instead of structures

→ It focuses on concepts instead of syntax.

→ It assumes interference is fundamental to understanding.

→ It introduces the idea of canonical meaning representation i.e. different words and structures represent same concept.
A language independent meaning representation.

→ Sentences are represented as a depicting actions using both abstract and real physical situation.

The primitive ACT's are building blocks.

ATRANS → transfer of an object / abstract relationship
eg - give

PTRANS → transfer of physical location of an object.
eg → go.

MTRANS → transfer of mental information
eg → tell.

MBUILD $\rightarrow$ Construction the new info from the new old one.

eg $\rightarrow$ Decide.

SPEAK $\rightarrow$ Utter a sound

eg $\rightarrow$ Say.

ATTEND $\rightarrow$ Focus a sense on stimulus
eg - listen, ~~watch~~ watch.

MOVE $\rightarrow$ movement of body part by the owner

eg $\rightarrow$ punch, kick.

PROPEL $\rightarrow$ Application of physical force to an object

eg $\rightarrow$ Push.

GRASP $\rightarrow$ Actor grasping an object.

eg $\rightarrow$ Clutch.

INGEST $\rightarrow$ Actor ingesting on an object

eg $\rightarrow$ eat drink

EXPEL $\rightarrow$ Actor getting rid of an object from body

eg $\rightarrow$ cry

PP $\rightarrow$ Pankaj Palagarism
 $\rightarrow$ Real world objects (picture produces)

ACT $\rightarrow$ Real world action

PA $\rightarrow$ Attributes of an object (picture aides)

AA $\rightarrow$ Attributes of an actions (Action aides)

T $\rightarrow$ Time

LOC $\rightarrow$ Location

I watched a match

I $\xleftrightarrow{P}$ MTRANS $\xleftarrow{O}$ Match $\xleftarrow{R}$ Conceptual processor
CP
eye

$\rightarrow$ Conceptual rules $\rightarrow$

Actor $\rightarrow$ performer of an ACT

ACT

OBJECT

RECIPIENT $\rightarrow$

DIRECTION $\rightarrow$

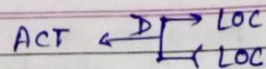
STATE

PP $\xleftrightarrow{}$ ACT

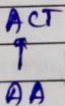
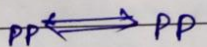
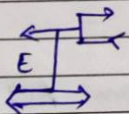
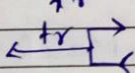
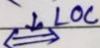
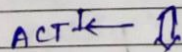
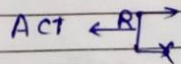
PP can perform action

PP $\leftrightarrow$ PA pp can described by an attribute

ACT $\xleftarrow{O}$ PP Acts can have an object



ACTs can have direction



17/04/2028

Conceptual Dependency

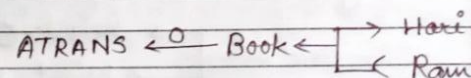
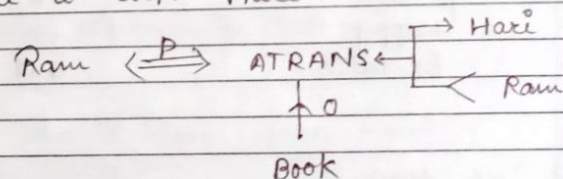
Some axioms / premisses

Conceptual links

past	P	Negation	/
future	f	conditional	C
present	nil	continuous	K

interrogative	?
time less	∞
start of transition	ts
End of transition	tf

Ram gave a book Hari



state of objects are described by

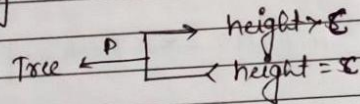
Fear Range (-10 to 0)

terrified	-9
scared	-5
anxious	-2
calm	0

Mental state Range (-10 to 10)

Catatonic	-9
Depressed	-5
upset	-3
sad	-2
OK	0
pleased	+2
Happy	+5
Ecstatic	+10

Tree grew



consciousness (0 to 10)

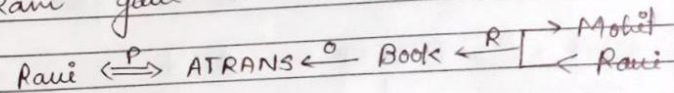
Unconscious	0
Asleep	5
awake	10
Highly drug consumed	>10

Some states take absolute value → length, color, Mass, Speed

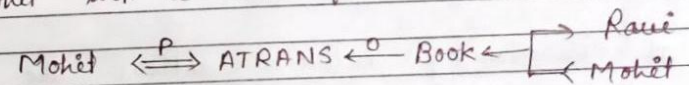
Some states are just relationship between objects

control
past
possession
ownership
contain
Proximity

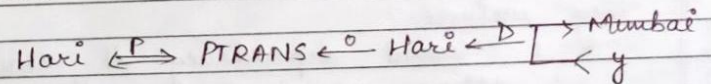
Ravi gave a book to Mohit.



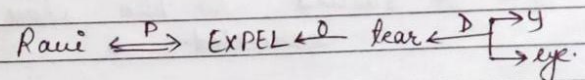
Mohit took a book from Ravi



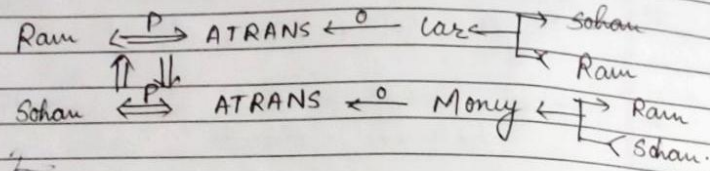
Hari went to Mumbai.



Ravi cried.



Ram sold his car to Sohan



Rohit annoyed Mohan

Rohit $\xleftrightarrow{P}$ Do

Mohan $\xleftrightarrow{P}$ Mental state(O)
 Mental state(O)

Rohit ate icecream with spoon.

Rohit $\xleftrightarrow{P}$ INGEST $\xleftarrow{O}$ Rohit
 Icecream Spoon

Relationship between actions are called dependencies

they include

1. casual dependency it means an action causes another actions.

eg $\rightarrow$ It rained so the class was cancelled.

② Temporal Dependency $\rightarrow$ It tells how the actions are related in time.

if class is over, fan will be

③ conditional dependency $\rightarrow$ one actions depends on another happening first.

if it rains we will not go for a picnic.

CD have following benefits $\rightarrow$

It makes representation language independent

It can be used in inter lingual representation.

It helps in disambiguation.

It helps in logical inference.

Limitations $\rightarrow$

① It is incomplete $\rightarrow$ Not all nua of human language can be captured by act alone.

② Complexity in Representation $\rightarrow$ As sentences become complex representing them accurately can require significant

computational resources.

- ③ lack of advanced concepts & some modern AI applications may require more sophisticated models that go beyond CD theory.

12-20